Alonso Perez-Lona

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Employment

Max Planck Institute for Mathematics in the Sciences, Leipzig Sep 2026 – present
Postdoctoral researcher in the Mathematical Structures in Physics group led by Daniel Roggenkamp.

Education

Ph.D. in Physics, Virginia Tech Aug 2021 – May 2026
Research on higher category-theoretic and homotopy structures arising in string and quantum field theory. Supervised by Eric Sharpe. CGPA 3.96.

B.S. Mathematics; B.S. Physics, New York University Abu Dhabi, UAE Aug 2017 – May 2021
Cum laude. CGPA 3.878.

International Baccalaureate Diploma, ITESM CEM, Mexico Aug 2014 – May 2017
Instituto Tecnológico y de Estudios Superiores de Monterrey. Score 39.

Publications

Topological interfaces between universes and generalized families of 2d QFTs 2026
With Pinak Banerjee, Yuhan Gai, Daniel Robbins, Subham Roy, and Eric Sharpe; appendix by Xingyang Yu.
Symmetries of 2d QFTs with topological point operators, which decompose into universes, described by multi-fusion categories; a classification of pairs of universes by the topological interfaces between them, and a proposed correspondence with generalized families of QFTs.
arXiv:2610.38680

Magnetic Higher-Form Symmetries as Dual Fundamental ∞-Groupoids 2026
A higher-geometric framework for magnetic higher-form symmetries of pre-quantum field theories, obtained by dualizing the shape of the stack of fields against a moduli stack of topological field theories, with an application to pure $\mathrm{Pin}^{\pm}(2)$ gauge theory.
arXiv:2610.03865

Notes on (-2)-form symmetries 2026
With Pinak Banerjee, Daniel Robbins, Subham Roy, Eric Sharpe, and Xingyang Yu.
A study of the nascent notion of (-2)-form symmetries in quantum field theory, using techniques from SymTFTs and category theory.
arXiv:2606.05543

Higher-form symmetries as higher automorphism bundles 2025
A cohesive higher topos-theoretic construction of higher-form global symmetries of field theories and their gaugings. Pure gauge theory of smooth 2-groups is analyzed explicitly, deriving in particular its (braided) center higher-form global symmetry. The general formulation is applied to the string 2-group $\mathrm{String}(G)$ of $G = SU(n)$, which describes $SU(n)$ gauge fields with vanishing instanton number, and to $G = U(1)$, which encodes the 5d supergravity bosonic fluxes, with attention to the interplay of higher-form global symmetries and nontrivial connections.
Accepted for publication in Adv. Theor. Math. Phys. | arXiv:2509.15301

Total instanton restriction via multiverse interference: Noncompact gauge theories and (-1)-form symmetries 2025
With Eric Sharpe, Xingyang Yu, and Hao Zhang.
The interplay between gauging certain (-1)-form symmetries, realized geometrically as gauge group extensions, and instanton restriction in two and four dimensions.
JHEP | doi:10.1007/JHEP05(2026)214 | arXiv:2508.00050

Categorified structures over moduli spaces: Anomalies, non-invertible symmetries, and exceptional holonomy 2025

With Eric Sharpe and Xingyang Yu.

Category-theoretic generalizations of symmetries of moduli spaces of CFTs, particularly those arising from string compactifications on exceptional holonomy manifolds.

arXiv:2506.19909

Anomaly resolution by non-invertible symmetries 2025

With Daniel Robbins, Subham Roy, Eric Sharpe, Thomas Vandermeulen, and Xingyang Yu.

Resolution of anomalous symmetries by extension to non-anomalous non-invertible symmetries, via SymTFTs and quasi-Hopf algebras.

Int. J. Mod. Phys. A | doi:10.1142/s0217751x25501611 | arXiv:2504.06333

Decomposition in 2d non-invertible gaugings 2025

Extension of the decomposition conjecture to certain non-faithfully acting non-invertible gauged symmetries in 2d QFT, including the corresponding decomposition partition functions.

J. Math. Phys. | doi:10.1063/5.0283628 | arXiv:2503.07727

Notes on gauging noninvertible symmetries, part 2: higher multiplicity cases 2024

With Daniel Robbins, Eric Sharpe, Thomas Vandermeulen, and Xingyang Yu.

Gauging of non-invertible symmetries in 2d QFT: non-multiplicity-free cases.

JHEP | doi:10.1007/JHEP05(2025)066 | arXiv:2408.16811

Discrete torsion in gauging non-invertible symmetries 2024

Extension of discrete torsion to generalized symmetries described by fusion categories, and its realization in pure gauge non-invertible actions on B fields.

J. Geom. Phys. | doi:10.1016/j.geomphys.2025.105423 | arXiv:2406.02676

Notes on gauging noninvertible symmetries, part 1: Multiplicity-free cases 2023

With Daniel Robbins, Eric Sharpe, Thomas Vandermeulen, and Xingyang Yu.

Gauging of non-invertible symmetries in 2d QFT.

JHEP | doi:10.1007/JHEP02(2024)154 | arXiv:2311.16230

Three-dimensional orbifolds by 2-groups 2023

With Eric Sharpe.

Generalization of discrete torsion and of decomposition to 3d orbifolds by 2-groups.

JHEP | doi:10.1007/JHEP08(2023)138 | arXiv:2303.16220

Research projects

Generalized symmetries and category-theoretic structures, Virginia Tech May 2022 – May 2026

Different perspectives on the interplay of generalized symmetries in quantum field theory and (higher) category theory.

Undergraduate capstone in mathematics, NYUAD Aug 2020 – May 2021

Higher category theory in topological quantum field theory (extended TQFT), and its application to Chern–Simons TQFT. Working with Hisham Sati, Urs Schreiber, and Qingtao Chen.

Undergraduate researcher in physics, NYU May 2020 – Aug 2020

Spontaneous breaking of center symmetry in $SU(N)$ Yang–Mills–Chern–Simons theory, using Dehn surgery, the Verlinde formula, and canonical quantization. Supervised by Sergei Dubovsky.

Undergraduate researcher in mathematics, NYUAD May 2019 – Dec 2019

Chern–Simons forms, computations, and generalizations, from a mathematical point of view. Supervised by Hisham Sati.

Leader of particle detector simulations, Arneodo Lab, NYUAD Aug 2017 – Nov 2018

Coordinated and produced the simulation of several detectors in Francesco Arneodo's lab, including a CubeSat.

Assistant in particle detector simulations, Institute of Nuclear Sciences, UNAM Jun 2016 – May 2017

Simulations for internal projects, mainly under the supervision of Eleazar Cuautle.

Talks, schools and conferences

Physical Mathematics of Quantum Field Theory, UMass Amherst Jun 2025

Participant, summer school on higher structures and QFT.

Southeastern Regional Mathematical String Theory Meeting, Duke University Nov 2024
Speaker: "Discrete torsion in gauging non-invertible symmetries."

Physical Mathematics of Quantum Field Theory, UMass Amherst Jun 2024
Participant, summer school on higher structures and QFT.

Geometric/Topological Quantum Field Theories and Cobordisms, NYU Abu Dhabi Mar 2023
Participant, conference on geometry and topology in QFT.

Physical Mathematics of Quantum Field Theory, UMass Amherst Aug 2022
Participant, summer school on higher structures and QFT.

Homotopical Methods of Fixed Point Theory, CU Boulder Jul 2022
Participant, summer school on fixed point theory and categorical duality.

Thematic Program in Field Theory and Topology, University of Notre Dame Jun 2022
Participant, summer school on applications of topology in QFT.

SENSE Detector School, Max Planck Institute, Germany Jun 2019
Three-day graduate-level school on low-light-level sensors.

SuperPhysMathics II Workshop, NYUAD May 2019
Workshop on supersymmetry and supergeometry.

2nd International Sharjah Spring School in Mathematics, UAE Mar 2019
Topology, geometry, and analysis.

Global Space Congress, UAE Mar 2019
Representative of the NYUAD CubeSat team.

Directed and independent study

Fellows program, Bradley Study Center Sep 2024 – May 2025
Fellows program of the Bradley Christian Center of Study.

Directed study in category theory, NYUAD Sep 2020 – Dec 2020
Basic Category Theory by T. Leinster in its entirety. Directed by Hisham Sati.

Independent study in Chern–Simons theory and gauge theory, NYU Feb 2020 – May 2020
Chern–Simons QFT (TASI lectures by G. Moore) and gauge theory (Rubakov, *Classical Theory of Gauge Fields*; lecture notes by D. Tong). Directed by Sergei Dubovsky.

Differential geometry II guided study, NYUAD Jan 2019 – May 2019
Fiber bundles and characteristic classes, with Hisham Sati.

Advanced directed study and research, ITESM CEM Jan 2016 – May 2016
Quantum mechanics and introduction to quantum computing. Directed by Francisco J. Delgado-Cepeda.

Leadership and community engagement

Organizer, String Theory Student Journal Club, Virginia Tech Aug 2022 – May 2026
Weekly in-depth presentations of technical topics in homotopy and category theory and string theory.

Founder, Theoretical Physics group, NYUAD Aug 2020 – May 2021
Study group preparing undergraduates for theoretical research, given the absence of theory at NYUAD; structured the work plan, including mathematics and graduate physics not offered on campus.

Tutor, accelerated physics course for the International Physics Olympiad, ITESM CEM Aug 2016 – Jul 2019
Created the course and tutored three students, one of whom won a bronze medal at the IPhO.

Co-founder and president, STEM projects group, ITESM CEM Jan 2016 – May 2017
Organized the group to create, fund, and develop STEM projects; principal evaluator of proposals; implemented hydroponics at the institute.

Mathematics tutor for elementary school students, Mexico City Aug 2016 – May 2017
Tutored fourth-year students from a disadvantaged area.

Awards and honors

- Clayton Williams Graduate Fellowship (2025–2026)
- Sigma Pi Sigma Physics Honor Society (2022)
- Phi Beta Kappa, NYU chapter (2021)
- NYU Founders Day Award (2021)
- Silver medal, Mexican Physics Olympiad (2016)
- Bronze medal, Mexican Physics Olympiad (2015)
- Gold medal, International Genetically Engineered Machine (iGEM) competition (2015)

Skills and languages

Computing: LaTeX (advanced), Geant4 (advanced), ROOT, Mathematica, C++, Java (intermediate), Python (beginner)
Languages: Spanish (native), English (proficient), French (advanced)